

A dietary supplement for female sexual dysfunction, Avlimil, stimulates the growth of estrogen-dependent breast tumors (MCF-7) implanted in ovariectomized athymic nude mice

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Abstract

Avlimil, a dietary supplement advertised to ameliorate female sexual dysfunction, is a mixture of eleven herbal components, and some herbal constituents of Avlimil (including black cohosh, licorice, red raspberry, red clover and kudzu) contain phenolic compounds, which are suggested to have estrogenic, anti-estrogenic, or androgenic potential for relieving menopausal symptoms. We hypothesize that Avlimil could modulate the growth of estrogen receptor positive human breast cancer (MCF-7) cells *in vitro* and *in vivo*. A dimethylsulfoxide (DMSO) extract of Avlimil (0.001–100 µg Avlimil powder equivalents/mL media) was tested for its estrogenic and anti-estrogenic effects on the growth of MCF-7 cells *in vitro*. We observed that the DMSO extract of Avlimil at low concentrations (0.1–50 µg/mL media) dose-dependently increased MCF-7 cell proliferation *in vitro*, and Avlimil DMSO extract at 100 µg/mL inhibited the growth of MCF-7 cells *in vitro*. Avlimil and some constituents (black cohosh and licorice roots) of Avlimil were fractionated by using sequential solvent extraction (hexane, ethyl acetate, and methanol) and the activities of the fractions were monitored by effects on the growth of MCF-7 cells. Depending on dosage (0.1–100 µg/mL media) both stimulatory and inhibitory effects of the extracts on the growth of MCF-7 cells were observed. The effect of dietary Avlimil at dosages approximating human intake was evaluated using ovariectomized mice implanted with MCF-7 cells. Animals were fed diets containing 500 ppm or 1000 ppm Avlimil for 16 weeks. Dietary Avlimil at 500 ppm stimulated MCF-7 tumors, but Avlimil at 1000 ppm had no apparent effect on the growth of MCF-7 tumors. The observation of stimulated tumor growth in the absence of uterine wet weight gains suggest that estrogenic/anti-estrogenic effects of Avlimil we observed may be dosage- and target tissue-specific and that Avlimil may not be safe for women with estrogen-dependent breast cancer. The different biological effects of fractionated Avlimil components and the different concentration dependencies warrant further compound identification and dose-response studies, especially at recommended intake levels that could have estrogenic effects in women.

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Keywords: Avlimil; Breast cancer; MCF-7; Black cohosh root; Licorice root; Isoflavones

Abbreviations: AIN93G, American Institute of Nutrition 93 growth semi-purified diet; BC, black cohosh; BCS, bovine calf serum; CD-BCS, charcoal-dextran stripped BCS; DMSO, dimethylsulfoxide; E₂, 17β-estradiol; ER, estrogen receptor; HRT, hormone replacement therapy; LC, licorice root; MCF-7, Michigan Cancer Foundation-7; MEM, minimal essential media; MTT, (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide); qRT-PCR, reverse transcription polymerase chain reaction; SE, standard error.

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1. Introduction

For women aged 40–55 breast cancer is the second leading cause of cancer death in the United States (Cancer.org, 2005a,b). However, most breast cancer cases (~77%) occur in postmenopausal women and most are estrogen receptor (ER)-positive (Harlan et al., 2002). Many breast cancer survivors experience sexual dysfunction (McKee and Schover, 2001), estimated at approximately 50% of long-term survivors (Ganz et al., 1998). Avlimil (also known as *Salvia rubus*) (Warner Health Care, 2003a,b), is the leading non-prescription dietary supplement for female sexual health, and has been on the market since January, 2003. In 1 year, the number of Avlimil customers reached 500,000 with the majority enrolled in a long-term preferred customer plan allowing for chronic exposure to these potentially estrogenic botanicals (Avlimil One Year Later, 2004). Avlimil is targeted to approximately 50 million women who have sexual problems for different reasons, including hormonal changes, menopause or aging processes where is the reference (CBS news, 2003 [‘women who have undergone hormonal fluctuations due to menopause or childbirth may find Avlimil helpful in restoring a sense of sexual well-being, without the use of hormone replacement therapy (HRT) or other prescription drugs’ Warner Health Care (2003a,b)]. The major herbal components in Avlimil are *Salvia officinalis* (sage leaf), *Rubus idaeus* (red raspberry leaf), *Pueraria montana* (kudzu root extract), *Trifolium pratense* (red clover extract), *Capsicum annuum* (capsicum pepper), *Glycyrrhiza glabra* (licorice root), *Morella cerifera* (bayberry fruit), *Turnera diffusa* (damiana leaf), *Valeriana officinalis* (valeriana root), *Zingiber officinale* (ginger root), and *Actaea racemosa* (black cohosh root). The flavones (apigenin, hispidulin, cirsimaritin) and abietane diterpenes from *S. officinalis* (sage leaf) act as benzodiazepine receptor-active components (Kavvas et al., 2003). Quinones (roylanone, 7 α -oxyroylanone, and 7 α -acetoxyroylanone) from *S. officinalis* have also been reported as bioactive compounds (antimicrobial and anti-inflammatory effects) (Spiridonov et al., 2003). *R. idaeus* (Red raspberry leaf) contains a variety of phenolic compounds and total phenolic contents for leaves are expressed as milligram of gallic acid equivalent per gram fresh weight (Wang and Lin, 2000), and its antioxidant effect has been documented (de Ancos et al., 2000; Kalt et al., 1999; Mullen et al., 2002; Wang and Lin, 2000). The presence of iso-flavones in Avlimil was indicated by the inclusion of red clover (Biochanin A and formonetin) and kudzu (puerarin), and their estrogen-like activities have been documented (Boue et al., 2003; Burdette et al., 2002). Although Avlimil does not contain estradiol, some plant materials [black cohosh root (*Cimicifuga racemosa*) and licorice root (*G. glabra*)], contain phenolic or terpenoid components that may have estrogenic and/or anti-estrogenic actions (Zava et al., 1998), and may modulate breast cancer in postmenopausal women. No information regarding consumption of Avlimil and possible safety issues, including

estrogen-dependent breast cancer, is available at the present time. It is also possible that this herbal mixture may show various biological effects at different dosages. Without a better understanding of its biological effect it is not possible to generalize that the use of this product is safe for postmenopausal women with breast cancer. In this study, we evaluated the potential estrogenic effects of Avlimil using well-described *in vitro* and *in vivo* assays to address this important women’s health concern.

2. Materials and methods

2.1. Chemicals and reagents

Avlimil was purchased from Warner Health Care, Inc. (Wallingford, CT). Minimal Essential Medium (MEM, without gentamicin, with glutamine) and phenol red-free MEM was purchased from the Media Facility at the University of Illinois at Urbana-Champaign. Bovine calf serum (BCS) was purchased from Hyclone (Logan, UT). Penicillin/streptomycin and trypsin/EDTA were purchased from Invitrogen (Houston, TX). Laboratory animal diet and dietary components were purchased from Dyets (Bethlehem, PA). Reagents for qRT-PCR were purchased from PE Applied Biosystems (Foster City, CA), Synthesgen (Houston, TX) and Invitrogen (Carlsbad, CA).

2.2. Maintenance of human breast cancer cells

MCF-7 cells are estrogen-dependent tumor cells isolated from a postmenopausal woman (Soule et al., 1973). MCF-7 cells were maintained in MEM supplemented with 5% BCS, 1% penicillin (100 U/mL)/streptomycin (100 μ g/mL), and 1 nM 17 β -estradiol (E₂). MCF-7 cells were maintained at 37 °C in a humidified atmosphere of 5% CO₂ in air as a monolayer culture in plastic culture plates (100 mm dia.). Seven days before the cell proliferation assay or the injection of MCF-7 cells into athymic mice, the media was switched to phenol red-free MEM containing 5% charcoal dextran stripped (CD)-BCS (Ju et al., 2001) and 1% pen/strep. MCF-7 in passages between 350 and 370, were used for cell assay and tumor growth study.

2.3. Preparation of extracts

2.3.1. DMSO Avlimil extract

To characterize the complex mixture of botanical products in Avlimil, we used a simple fractionation method that minimizes degradation and allows for systematic evaluation of the extracts using a cell growth assay. The purple film coat of Avlimil tablet was removed using a razor blade. The naked tablet was ground to a fine powder using a mortar and pestle. The Avlimil powder was mixed in DMSO (1 g in 10 mL) in a centrifuge tube, vortexed vigorously, incubated at 37 °C for 30 min, and centrifuged for 10 min at 1000 rpm at 24 °C. The supernatant portion, containing the equivalent of 100 μ g of the original Avlimil powder/ μ L, was recovered, filter-sterilized (0.45 μ m), aliquoted, stored at –20 °C, and used for *in vitro* cell assay. The insoluble material remaining was discarded.

Bioactive material bound to the filter and the insoluble materials that may have activity were not assayed due to inherent technical problems associated with this type of experimentation.

2.3.2. Sequential hexane, ethyl acetate and methanol Avlimil extracts

The Avlimil powder was mixed with hexane (1 g/mL) in a glass tube, vortexed vigorously, incubated at 37 °C for 30 min, and centrifuged for 10 min at 1000 rpm at 24 °C. The supernatant was decanted and dried using nitrogen gas, reconstituted with DMSO as above (100 μ g Avlimil powder equivalents/ μ L), filter-sterilized, aliquoted, stored at –20 °C, and used for *in vitro* cell assays. The resulting solids portion was air-dried, and mixed with ethyl acetate (1 g/mL) in a glass tube, vortexed vigorously,

incubated at 37 °C for 30 min, and centrifuged for 10 min at 1000 rpm at 24 °C. The supernatant was decanted and dried using nitrogen gas, reconstituted with DMSO (100 µg/µL), filter-sterilized, aliquoted, stored at –20 °C, and used for *in vitro* cell assays. The resulting solids portion was air-dried, and mixed with methanol (1 g/mL) in a glass tube, vortexed vigorously, incubated at 37 °C for 30 min, and centrifuged for 10 min at 1000 rpm at 24 °C. The supernatant was decanted and dried using nitrogen gas, reconstituted with DMSO (100 µg/µL), filter-sterilized, aliquoted, stored at –20 °C, and used for *in vitro* cell assays. The insoluble material remaining was discarded. To reduce possible batch to batch variability in Avlimil composition, we purchased a large quantity of Avlimil sufficient for all cell assays and feeding studies, and extracts were prepared at one time.

2.3.3. Black cohosh root and licorice root supplements extracts

We purchased black cohosh root supplement from Nature's Resource (currently Nature Made, Mission Hills, CA) and licorice root supplement from Nature's Way (Springville, Utah). A capsule was removed and the contained powder was used to prepare extracts. We used the described sequential solvent extraction method to separate the different constituent phytochemicals from black cohosh root or licorice root supplements.

2.4. Cell viability assay

To evaluate the effect of Avlimil on MCF-7 cell growth, a modified colorimetric MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assay (Ju et al., 2004) was used. The MTT assay is a widely used screening method to measure cell viability and proliferation by measuring mitochondrial dehydrogenase activity. Because of potential false positive results due to possible interference of natural substances in the extracts and MTT (Bruggisser et al., 2002), three sets of results obtained by MTT assay were compared with ones by manual counting, and no significant difference was observed (data not shown). MCF-7 cells were grown in estrogen-free culture media (phenol red-free MEM containing 5% CD-BCS and 1% pen/strep) for two weeks prior to the treatment. For the growth experiments, MCF-7 cells (1.5×10^4) were inoculated in triplicate in 1 mL of estrogen-free culture media in a 24-well polystyrene culture plate. After 24 h the MCF-7 cells in each well were washed with 1 mL of phosphate buffered saline (150 mM PBS, pH 7.4, without Ca^{2+} and Mg^{2+}), and treated with various concentration of Avlimil (0.001–100 µg/mL media), black cohosh (0.1–100 µg/mL media), or licorice (0.1–100 µg/mL media) extracts every 48 h. Control with the solvent, DMSO ($\leq 0.1\%$, v/v), was also tested. After the 5-day treatment, cells were treated with MTT for 5 h and treated with 10% SDS (in 0.01 N HCl) for 12–18 h. Optical density (OD) was measured at 570 nm. OD values were normalized to number of cells based on a standard curve. Each set of MTT assays was repeated three times. To reduce the possible batch to batch variations in black cohosh and licorice supplements, a single large quantity of each supplement was purchased and extracts were prepared at one time.

2.5. Athymic nude mice

Female athymic BALB/c (nude) mice were purchased from Charles River Laboratories (Wilmington, MA) and acclimated for a week.

Mice were ovariectomized at 21-d of age by the vendor and then allowed to recover for 7 d. During the study, the mice were maintained under the standard light/dark cycle (12 h light, 12 h dark).

2.6. E₂ pellet preparation

An E₂ pellet contained 2 mg of E₂ and 18 mg of cholesterol. An E₂ pellet was placed subcutaneously under the skin of each mouse before MCF-7 cells were transplanted into the mice. This dosage of E₂ pellet will produce pharmacological-physiological levels of E₂ of approximately 2–3 nM in plasma and induce maximal tumor growth (Ju et al., 2006).

2.7. Diet formulation

American Institute of Nutrition 93 growth diet (AIN93G) semi-purified diet (Dyets, Bethlehem, PA) was selected as a base diet for control mice as it has been established as meeting all of the nutritional requirements of mice (Reeves, 1997). Soy oil was substituted with corn oil in order to remove other soybean components in the diet. Treatment animals were fed AIN93G diet plus different dosages of ground Avlimil (500 or 1000 ppm).

2.8. Analysis of tumor growth

Three days after the E₂ pellets were inserted into mice, MCF-7 cells [1×10^5 cells per 40 µl of Matrigel® (Collaborative Biomedical Products, Bedford, MA)] were injected at 40 µl per site into each of the four spots on the backs of the athymic mice. All mice were maintained on AIN93G diet. Two weeks after E₂ pellet implantation, the average tumor surface area reached 13.76 mm². At this point, mice were divided into four treatment groups (5–6 mice/group); MCF-7, E₂, 500 ppm Avlimil, and 1000 ppm Avlimil groups. E₂ pellets were removed from all mice except the E₂ group, and dietary Avlimil treatment was started. Tumor growth and body weight were measured weekly, and cross-sectional area was determined using the formula $[\text{length}/2 \times \text{width}/2 \times \pi]$ (Gottardis et al., 1989). Food intake was measured throughout the study. At the end of the study, uterine wet weight was measured. Tumors and blood samples were collected for analysis.

2.9. Expression of Ki-67 (a measure of proliferation in tumors)

Cell proliferation in tumors was determined using immunohistochemical analysis. Tumors in 10% formalin were embedded in paraffin blocks, cut into 5 µm sections and placed on microscope slides. Slides were deparaffinized by immersing in xylene twice for 12 min each and hydrated by immersing in a series of 100% ethanol, 95% ethanol, and three times in dH₂O for 5 min each. To block endogenous peroxidase, slides were immersed in 0.3% H₂O₂ for 20 min and then washed with dH₂O. Slides were placed in citrate buffer (pH 6), microwaved for 20 min, and then cooled at 24 °C. Then, slides were washed in PBS (pH 7.4) for 5 min. Anti-human Ki-67 antibody (PharMingen, San Diego, CA) (1:3000 dilution in 1%BSA/PBS) was added to the slides, which were then incubated for 1 h at 24 °C in a humidity chamber. Slides were washed in PBS before the addition of biotinylated anti-mouse secondary antibody (VECTASTAIN Elite ABS reagent, Vector Laboratories, Inc. Burlingame, CA) was added to slides and incubated for 30 min at 24 °C. Slides were then washed in PBS. One drop of 3,3'-diaminobenzidine tetrahydrochloride (DAB) prepared right before use, was added to each tissue section on the slide allowed for 10 min, counterstained with 20% hematoxylin for 5 min after washing with dH₂O and PBS, and counterstained with 20% hematoxylin for 1 min. The slides were then dehydrated by submerging them in 80% ethanol for 5 min, 95% ethanol for 5 min, and 100% ethanol for 5 min, followed by submerging slides in xylene four times for 5 min each. Slides were mounted and analyzed using a light microscope. Both positive and background stained cells were counted in a given area of tissue. A total of 25 fields from 5 tumors per treatment group were evaluated. The data were then presented as a percentage of cells proliferating in a given area of tumor.

2.10. RNA preparation and qRT-PCR

Avlimil-induced pS2 mRNA expression in MCF-7 cells and MCF-7 tumors was analyzed using a qRT-PCR. The pS2 gene is an estrogen-responsive gene, which is expressed in breast cancer cells, but not in normal mammary cells (Rio et al., 1987). The pS2 gene provides an ideal model system to study estrogen-regulated gene expression, and expression of the pS2 gene has been used as a marker of estrogen responsiveness in estrogen receptor-containing breast cancer cells, an indicator of disease progression (Schwartz et al., 1991). MCF-7 cells ($\sim 2 \times 10^6$ cells), and tumors (≤ 30 mg) with areas similar to the mean tumor surface area of the

each treatment groups were used for mRNA analysis using RNeasy® Mini kit (Qiagen Inc., Valencia, CA). cDNA were generated using 10 ng of RNA and TaqMan Reverse Transcription Reagents (PE Applied Biosystems, Foster City, CA). The pS2 primers and fluorescence (6-FAM)-labeled probes were designed using Primer and Probe Design Express (PE Applied Biosystems) (Ju et al., 2004). The human GAPDH primers and a fluorescent (6-FAM/TAMRA)-labeled probe (User Bulletin #2, PE Applied Biosystems) were used as a control. PCR and analysis of PCR products were performed using the ABI PRISM 7700 Sequence Detector (PE Applied Biosystems). Data were analyzed using a comparative threshold cycle (Ct) method (User Bulletin, PE Applied Biosystems). Amplicons were run as triplicates in separate tubes to permit quantification of target genes normalized to a control, GAPDH.

2.11. Plasma isoflavone levels

Blood samples were collected by cardiac puncture and placed into EDTA-containing tubes and centrifuged at $500 \times g$ for 5 min. Plasma samples were stored at -20°C until analyzed. Avlimil contains isoflavones from red clover (Sivesind and Seguin, 2005) and kudzu root (Zhang et al., 2005). The total genistein, daidzein, and equol levels in plasma were analyzed by liquid chromatography and isotope dilution electrospray tandem mass spectrometry (LC–ES/MS/MS) using a modification of the method previously described (Twaddle et al., 2002).

2.12. Statistics

Data from *in vitro* cell proliferation assay, tumor area at final week, qRT-PCR, IHC analysis and uterine wet weight were analyzed accordingly using one-way or repeated-measures analysis of variance according to the characteristics of the data set using the SAS program. If the overall treatment *F*-ratio was significant ($p < 0.05$), the differences between treatment means were tested with Fisher's least significant difference (LSD) test.

3. Results

3.1. Avlimil (DMSO extract) stimulates the growth of MCF-7 cells *in vitro*

A DMSO extract of Avlimil significantly increased cell proliferation at greater than $0.1 \mu\text{g/mL}$ media, relative to the MCF-7 control, respectively ($p < 0.05$) (Fig. 1A). Avlimil extract at concentrations lower than $10 \mu\text{g/mL}$ media, did not change 1 nM E_2 -stimulated MCF-7 cell growth, while concentrations greater than $50 \mu\text{g/mL}$ media significantly inhibited 1 nM E_2 -induced MCF-7 cell growth (Fig. 1B). The cell growth induced by Avlimil was blocked by ICI 182,780 (100 nM) treatment (Fig. 1C).

3.2. Ethyl acetate and methanol extracts of Avlimil shows the more potent stimulatory effect than the hexane extract on the growth of MCF-7 cells *in vitro*

A hexane extract ($\geq 1 \mu\text{g/mL}$ media), an ethyl acetate extract ($\geq 0.1 \mu\text{g/mL}$), and a methanol extract ($\geq 10 \mu\text{g/mL}$) increased the growth of MCF-7 cells (Fig. 2).

3.3. Effects of black cohosh extracts on the growth of MCF-7 cells *in vitro*

In the absence of E_2 , a DMSO extract and a hexane extract of black cohosh ($\geq 50 \mu\text{g/mL}$) had inhibitory

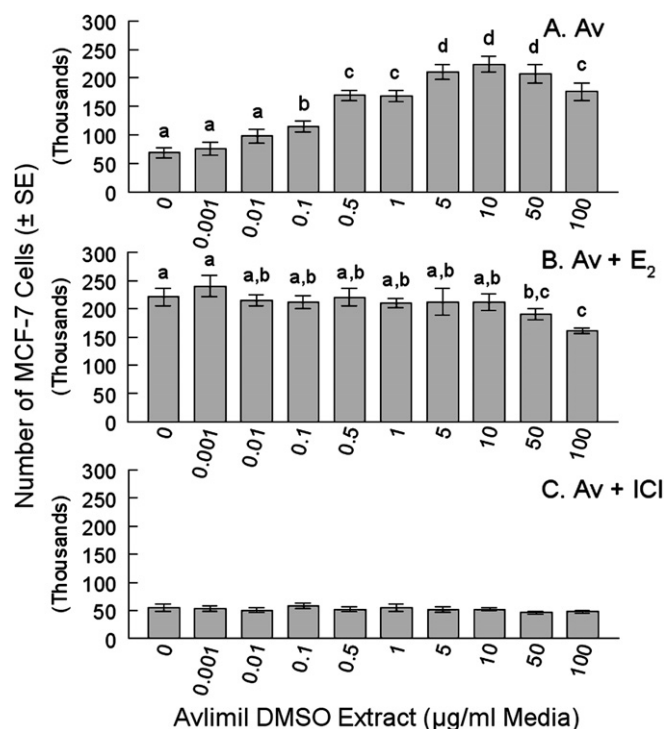


Fig. 1. Effect of a DMSO extract of Avlimil on the growth of MCF-7 cells *in vitro*. DMSO extract of Avlimil (A), DMSO extract of Avlimil + 1 nM E_2 (B) and DMSO extract of Avlimil + 100 nM ICI 182,780 (C) at concentrations between 0.001 and $100 \mu\text{g/mL}$ media was evaluated for its effect on MCF-7 cell growth using MTT assay. MCF-7 cellular proliferation is expressed as mean number of MCF-7 cells $\pm$ SE. Bars with different letter are significantly different, $p < 0.05$. A dose-dependent biphasic effect of DMSO extracts of Avlimil on the growth of MCF-7 cells was observed (stimulatory at 0.1 – $50 \mu\text{g/mL}$ and inhibitory at $100 \mu\text{g/mL}$) (A). An anti-estrogenic effect of DMSO extracts of Avlimil at high doses (50 – $100 \mu\text{g/mL}$) on the growth of MCF-7 cells in the presence of a stimulatory dose of E_2 (1 nM) (B). The stimulatory effect of a DMSO extract of Avlimil was blocked by the anti-estrogen, ICI 182,780 (C).

effects, an ethyl acetate extract had no effect, and a methanol extract ($\geq 0.5 \mu\text{g/mL}$) had a stimulatory effect on the growth of MCF-7 cells (Fig. 3, white bars). In the presence of E_2 , the DMSO extract ($\geq 0.5 \mu\text{g/mL}$), the hexane extract ($\geq 1 \mu\text{g/mL}$), the ethyl acetate extract ($\geq 5 \mu\text{g/mL}$), and the methanol extract (at $100 \mu\text{g/mL}$) had inhibitory effects on the E_2 -stimulated growth of MCF-7 cells (Fig. 3, gray bars).

3.4. Effects of licorice extracts on the growth of MCF-7 cells *in vitro*

In the absence of E_2 , a DMSO extract of licorice ($\geq 5 \mu\text{g/mL}$), a hexane extract ($\geq 1 \mu\text{g/mL}$), an ethyl acetate extract ($\geq 0.1 \mu\text{g/mL}$), and a methanol extract ($\geq 0.1 \mu\text{g/mL}$) had stimulatory effects on the growth of MCF-7 cells (Fig. 4, white bars). In the presence of E_2 , the DMSO extract ($\geq 0.1 \mu\text{g/mL}$), the hexane extract ($\geq 5 \mu\text{g/mL}$), the ethyl acetate extract ($\geq 5 \mu\text{g/mL}$), and the methanol extract ($\geq 50 \mu\text{g/mL}$) had inhibitory effects

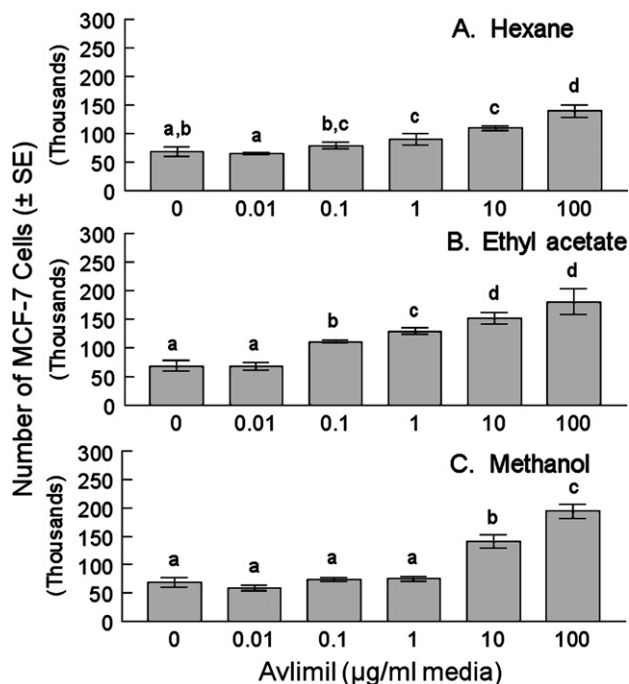


Fig. 2. Effect of hexane (A), ethyl acetate (B), and methanol (C) extracts of Avlimil at concentrations between 0.1 and 100 µg/mL media was evaluated for its effect on MCF-7 cell growth using MTT assay.

on the growth of E₂-stimulated MCF-7 cells (Fig. 4, gray bars).

3.5. Avlimil DMSO extract increased pS2 mRNA expression in MCF-7 cells in vitro

To evaluate the capability of Avlimil to modulate the expression of pS2 mRNAs, an estrogen-responsive gene, MCF-7 cells were treated with the Avlimil DMSO extract at various concentrations and analyzed using qRT-PCR. The DMSO extract of Avlimil at 1 and 10 µg/mL, significantly increased pS2 mRNA expression in MCF-7 cells, whereas the extract at 100 µg/mL, decreased pS2 expression (Fig. 5). pS2 expression is presented as relative pS2 mRNA expression ± SE. E₂ (1 nM) enhanced pS2 expression by 6.6 ± 0.6 -fold over the MCF-7 cell control (0 control on Fig. 5) ($p < 0.05$).

3.6. Effect of dietary Avlimil intake on MCF-7 tumor growth in athymic mice

We selected 500 and 1000 parts per million (p.p.m.) dietary dosages to mimic the dietary exposure in women. Our assumptions are that a woman consumes 0.5–1.5 kg of food per day. In women, one 756 mg tablet per day would be approximately 500–1500 ppm. We selected two dosages at the lower end of this range for this current study. It is important to point out that if a woman was to consume food similar to the mouse diet, she would consume at the low end of this range and the dosages of 500 and 1000 ppm utilized in this investigation are conservative. Thirteen weeks after the E₂ (2 mg) implantation, average tumor surface area in

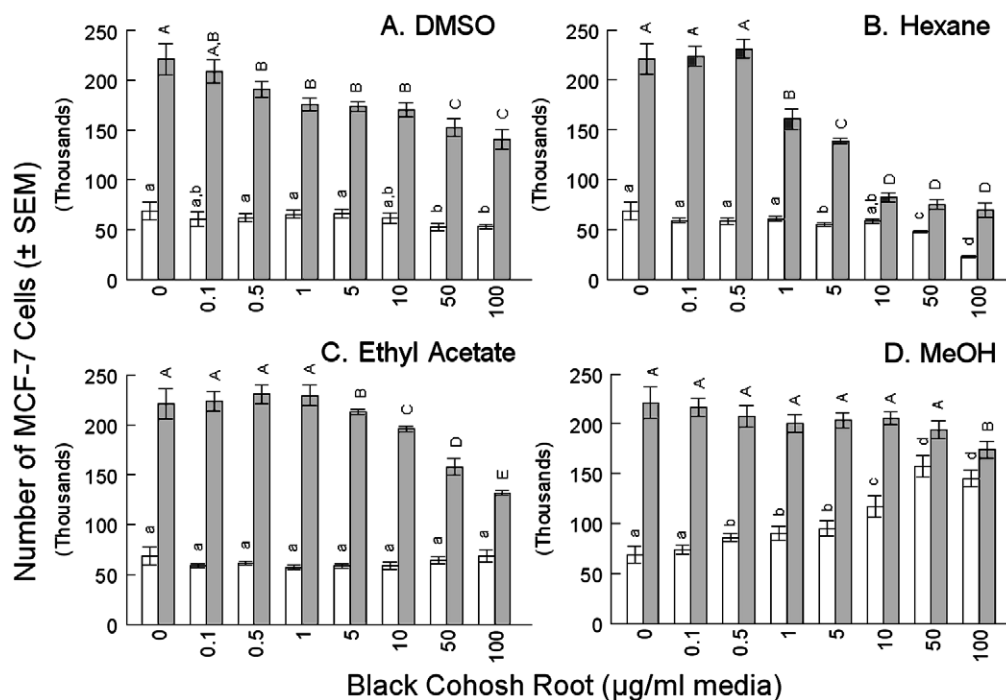


Fig. 3. Effects of black cohosh root extracts on the growth of MCF-7 cells *in vitro*. Effect of DMSO (A), hexane (B), ethyl acetate (C), and methanol (D) extracts of black cohosh root supplement at concentrations between 0.1 and 100 µg/mL media was evaluated for its effect on MCF-7 cell growth using MTT assay. (Grey Bars: Extracts + 1 nM E₂, Clear Bars: Extracts).

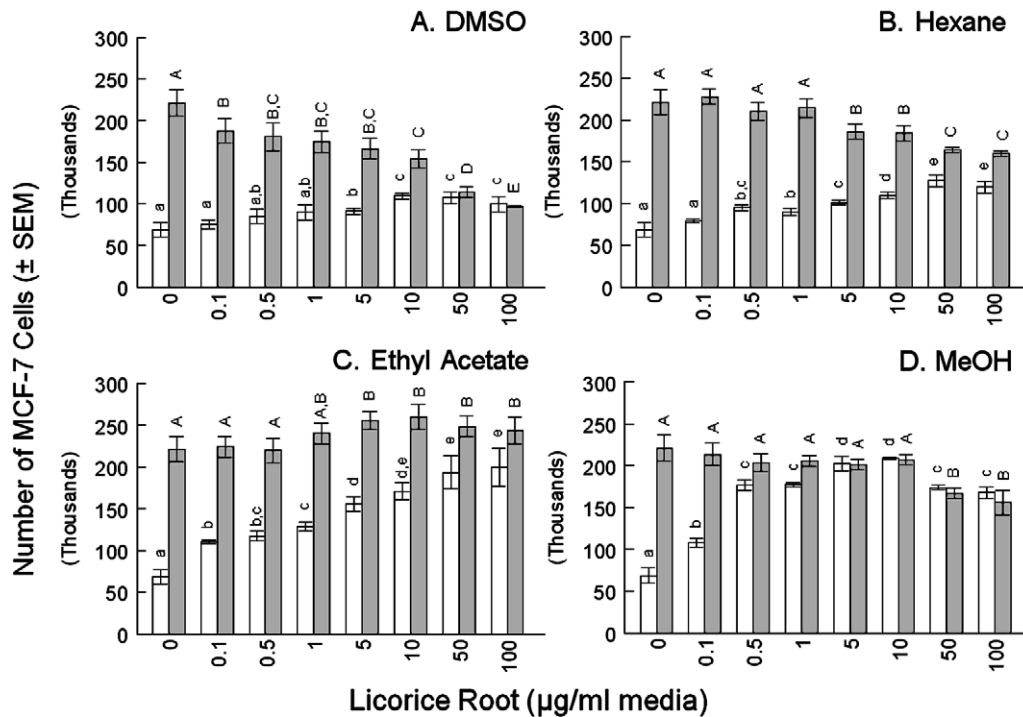


Fig. 4. Effects of licorice root extracts on the growth of MCF-7 cells *in vitro*. Effect of DMSO (A), hexane (B), ethyl acetate (C), and methanol (D) extracts of licorice root supplement at concentrations between 0.1 and 100 μ g/mL media was evaluated for its effect on MCF-7 cell growth using MTT assay. (Grey Bars: Extracts + 1 nM E_2 , Clear Bars: Extracts).

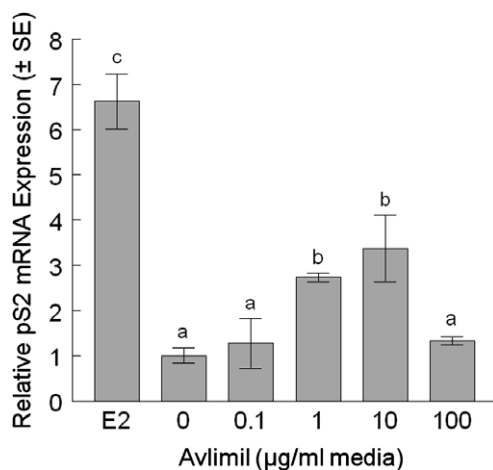


Fig. 5. Effect of the DMSO extract of Avlimil on pS2 mRNA expression in MCF-7 cells. MCF-7 cells were treated with DMSO extract of Avlimil for 24 h, and analyzed using qRT-PCR. Numbers on the Y-axis represent the relative mRNA level; bars $\pm$ SE. Data points were obtained from three replicates. GAPDH was used as a standard. Bars with different letter are significantly different, $p < 0.05$. DMSO extracts of Avlimil produced a biphasic effect on the expression of the estrogen responsive gene marker, pS2 mRNA (stimulatory at 1 and 10 μ g/mL and inhibitory at 100 μ g/mL) in MCF-7 cells.

the E_2 group reached 105.2 ± 11.1 mm² (Fig. 6), and mice in the E_2 group were terminated due to tumor burden. Tumors in the rest of the treatment groups were monitored five more weeks when average tumor size of the 500 ppm Avlimil

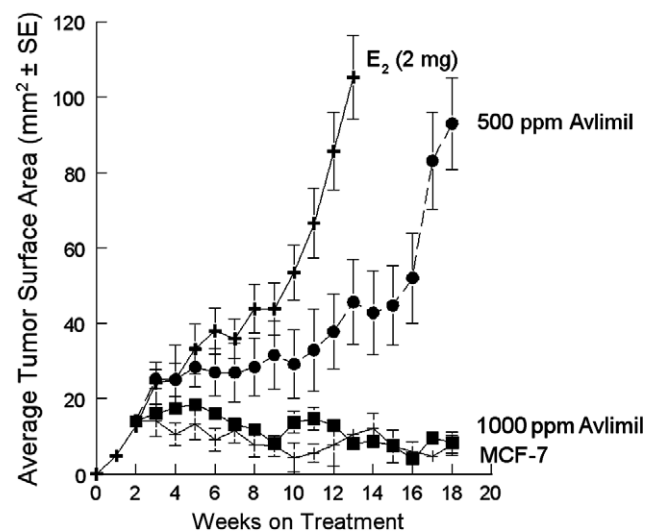


Fig. 6. Effect of Dietary Avlimil on the growth of MCF-7 tumors implanted in ovariectomized athymic mice. Two weeks after the E_2 (2 mg pellet) implantation the average tumor size reached 18 mm². At this time, E_2 pellets were removed from all of the mice except the E_2 control group and mice were assigned into four treatment groups [E_2 (5 mice; $n = 17$ tumors), MCF-7 that were fed AIN 93 G diet alone (5 mice; $n = 7$ tumors), 500 ppm Avlimil (6 mice; $n = 18$ tumors), and 1000 ppm Avlimil (6 mice; $n = 8$ tumors)], and mice were started on the experimental diets (week 2). Tumors were then measured weekly for 16 weeks (11 weeks for the E_2 group). Data are expressed as average cross sectional tumor area (mm² $\pm$ SE) for all tumors in each treatment. Dietary Avlimil induced a biphasic effect on MCF-7 tumor growth in athymic nude mice (stimulatory at 500 ppm and no effect at 1000 ppm).

group reached $92.9 \pm 12.1 \text{ mm}^2$ (Fig. 6). Sixteen weeks (from week 2 to week 18 in Fig. 4) of 500 ppm Avlimil treatment stimulated tumor growth by $12.1\times$ over the MCF-7 control group ($p < 0.05$). After the removal of E_2 pellets at week 2, tumor sizes of the MCF-7 control group and the 1000 ppm Avlimil group were regressed to $7.7 \pm 2.7 \text{ mm}^2$ and $8.3 \pm 2.8 \text{ mm}^2$, respectively, and there was no statistical difference between the MCF-7 control and 1000 ppm Avlimil groups ($p = 0.8590$).

3.7. Cell proliferation in tumors

Percentage of Ki-67 expression in MCF-7 tumors was $2.4 \pm 0.7\%$. E_2 treatment increased the percentage of cell proliferation to $55.1 \pm 7.2\%$ ($p < 0.05$), and 500 ppm Avlimil also increased cell proliferation to $42.4 \pm 6.0\%$ ($p < 0.05$) but 1000 ppm Avlimil produced no effect ($2.1 \pm 1.1\%$, $p = 0.9633$) relative to MCF-7 (Fig. 7).

3.8. pS2 mRNA expression in tumors

Dietary Avlimil-induced pS2 mRNA expression in tumors was analyzed using qRT-PCR. pS2 expression is presented as relative pS2 mRNA expression $\pm$ SEM. E_2 increased relative pS2 mRNA expression levels in tumors by $4.8 \pm 1.0\times$ over the MCF-7 group ($p < 0.05$), 500 ppm Avlimil by $3.1 \pm 0.9\times$ ($p < 0.05$); 1000 ppm Avlimil reduced pS2 expression to $0.9 \pm 0.2\times$ ($p = 0.8149$) (Fig. 8). There was no statistical difference in pS2 expression between E_2 and 500 ppm Avlimil groups ($p = 0.0891$).

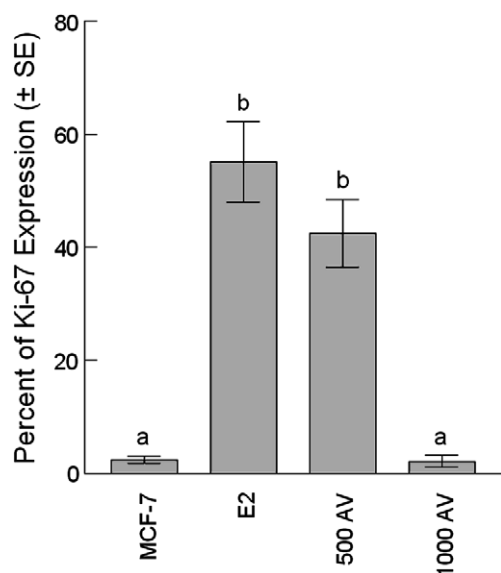


Fig. 7. Effect of dietary Avlimil on the cellular proliferation on MCF-7 tumors. Ki-67 expression was evaluated as a measure of proliferation. A total of 25 fields from 5 tumors per treatment group were evaluated. Y-axis (cell proliferation) is presented as the percentage of cells actively proliferating in a given area of tissue ($\pm$ SEM). Bars with different letters differ, $p < 0.05$. Dietary Avlimil induced a biphasic effect on the expression of proliferation marker ki-67 in tumors (stimulatory at 500 ppm and no effect at 1000 ppm).

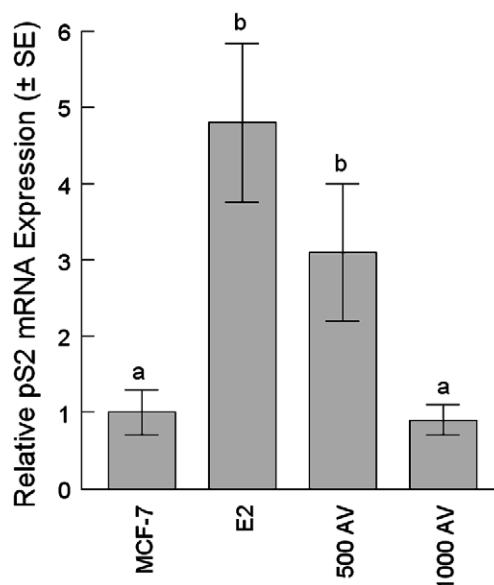


Fig. 8. pS2 mRNA expression in tumors. Six to seven tumors per treatment group were analyzed using qRT-PCR. Numbers on the Y-axis represent the relative mRNA level; bars $\pm$ SEM. GAPDH was used as a standard. Bars with different letter are significantly different, $p < 0.05$. Dietary Avlimil induced a biphasic effect on the expression of pS2 mRNA in tumors (stimulatory at 500 ppm and no effect at 1000 ppm).

3.9. Uterine wet weight, body weight, and food intake

Average uterine wet weights of mice in the MCF-7, E_2 , 500 ppm Avlimil, and 1000 ppm Avlimil groups were $24.3 \pm 5.2 \text{ mg}$, $101.75 \pm 9.1 \text{ mg}$ ($p < 0.05$), $17.0 \pm 2.2 \text{ mg}$ ($p = 0.1018$), and $17.3 \pm 2.2 \text{ mg}$ ($p = 0.1158$), respectively (Fig. 9). E_2 implantation increased uterine wet weight

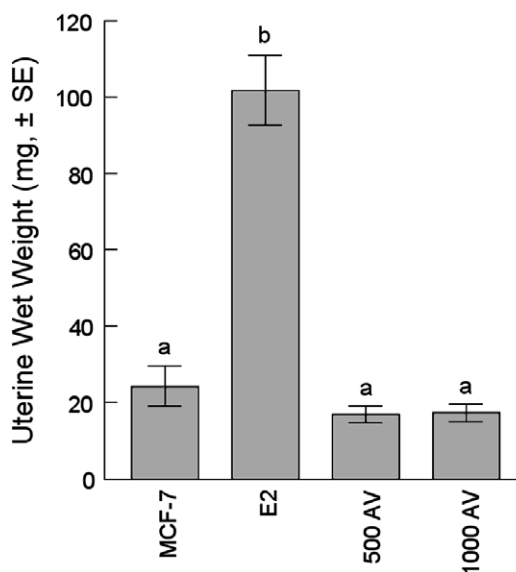


Fig. 9. Effect of dietary Avlimil on uterine wet weight. Data are expressed as average uterine wet weight (mg $\pm$ SEM) for all uteri in each treatment. Bars with different letter are significantly different, $p < 0.05$. Dietary Avlimil did not induce uterotrophic effect.

by 3.6× over the MCF-7 group. No uterotrophic effect was observed in any of the Avlimil groups. Body weight was also monitored weekly and no significant difference was observed among the treated and control groups (data not shown). No significant difference was observed in food intake among any of the treatment groups (data not shown).

3.10. Total isoflavone levels in plasma

Total genistein, daidzein, and equol levels in plasma were found to be less than the detection limit (0.005 μM , data not shown).

4. Discussion

Approximately 30% of Americans are using at least one form of complementary and alternative medicine (CAM) per year, typically for chronic rather than life-threatening medical conditions (Eisenberg et al., 1993). A recent review on CAM in early breast cancer showed that 45% of breast cancer patients used CAM (Gerber et al., 2006). Consumption of dietary supplements and herbal supplements are considered as CAM utilization (Ness et al., 2005). There is great public interest in alternative medicines and the use of herbal remedies for overall well-being. Herbal medicine is based on the premise that plants contain natural substances that can promote health and alleviate illness. The active ingredients, optimal dosages, short- and long-term effects, and contra-indications of the herbal substances have not been studied in most cases, and the efficacy and the safety of these products are generally not known. Avlimil is one of these herbal products, and its consumption has rapidly increased. Recently, the manufacturer introduced 'Avlimil Complete' (Avlimil + soy isoflavones) with a claim 'to restore a woman's natural balance without the use of steroids or drugs' Avlimil Complete (2004). The potential for botanical remedies to counter medical problems should not be underestimated, although no clinical trials have shown Avlimil to have beneficial effects on female sensuality. However, safety of these products has not been adequately evaluated regarding increased breast cancer risk. In this study, we found evidence for estrogenic stimulation of estrogen-dependent breast cancer by Avlimil. We demonstrated *in vitro* that the DMSO extract of Avlimil increased estrogen-dependent breast cancer cell proliferation in a dose-dependent manner and that Avlimil DMSO extract causes cellular proliferation at concentrations as low as 0.01 $\mu\text{g/mL}$ media *in vitro* (Fig. 1a). Next, we evaluated the interaction of Avlimil DMSO extract with E_2 and demonstrated that at higher concentrations the Avlimil DMSO extract inhibited E_2 -stimulated MCF-7 cell growth, suggesting anti-estrogenic activity (Fig. 1b). An anti-estrogen, ICI 182,780, blocked the Avlimil extract-stimulated MCF-7 cell growth (Fig. 1c), which suggested that the stimulatory effect we observed (Fig. 1a) is due to an ER-mediated mechanism(s). Avlimil DMSO

extract had no effect on the growth of estrogen-independent breast cancer cells (MDA-MB-231) (data not shown) supporting the notion that the cellular proliferative effect of Avlimil DMSO extract observed in MCF-7 cells (Fig. 1) is via ER-mediated mechanisms. Avlimil is an herbal mixture, therefore, we speculated that sequential chemical extraction using solvents with different polarity would fractionate the mixture into bioactive components. We compared various extracts of Avlimil (hexane, ethyl acetate, and methanol) based on the ability to alter growth of MCF-7 cells. The ethyl acetate and methanol extracts had more stimulatory effect on MCF-7 cell growth compared to the hexane extract (Fig. 2). Then we selected two components of Avlimil (black cohosh and licorice roots) that may confer estrogenic (or anti-estrogenic) activity. We used commercially available black cohosh and licorice dietary supplements because these are also widely available and consumed by women. We compared effects of DMSO, hexane, ethyl acetate, and methanol extracts of black cohosh (Fig. 3) and licorice (Fig. 4) on the growth of MCF-7 cells in the presence and in the absence of E_2 . These results suggest that multiple constituents of Avlimil can either stimulate the growth of ER-positive breast cancer cells or inhibit the E_2 -induced growth of breast cancer cells. The results from two selected components of Avlimil, black cohosh root and licorice root extracts, further confirmed the complex nature of these herbal products where both stimulatory and inhibitory fractions were observed. Inhibitory effect from ethanol and iso-propanol extracts of black cohosh on the growth of ER-positive and -negative breast cancer cells have been reported previously (Bodinet and Freudenstein, 2002; Hostanska et al., 2004; Zierau et al., 2002). Estrogen-like activities of licorice constituents in vascular tissues and osteoblasts were also reported (Somjen et al., 2004a,b). Estrogenic and anti-estrogenic activities of isoliquiritigenin, a chalcone constituent of licorice extract, on the growth of MCF-7 have been demonstrated and it was speculated that physiological concentrations exerted estrogenic effects through transactivation of $\text{ER}\alpha$, whereas high levels induced cytotoxicity through different mechanisms (Maggiolini et al., 2002) as observed with other phytoestrogens (Akiyama et al., 1987; Barkhem et al., 1998; Kuiper et al., 1998; Maggiolini et al., 2001; Walker et al., 2000). The effects on MCF-7 cells in this study may not be solely attributed to phenolic compounds but also triterpene glucoside fractions of black cohosh. Thus, to the observed effects could be caused by pleiotropic actions of individual components or synergism between compounds (Jo et al., 2005; Stromeier et al., 2005).

We confirmed that the stimulation of cell growth by intermediate concentrations of Avlimil extract was due to its estrogen agonist action by measuring elevated expression of the estrogen-regulated gene, pS2; however, the highest concentration reversed the stimulation back to control levels suggesting the action of an inhibitory component (Fig. 5). We also evaluated the effect of dietary Avlimil on the growth of MCF-7 tumors *in vivo* (Fig. 6) using a mouse

xenograft model. This model has been used extensively for several decades for pre-clinical investigations of breast cancer treatment and safety regarding stimulation of tumor growth (Dowsett et al., 1996; Gottardis et al., 1989; Jordan et al., 1989; Ju et al., 2001, 2002; Santen et al., 2001). The model, which uses ovariectomized mice implanted with MCF-7 tumor cells, is relevant to older women because the low circulating level of E_2 mimics what is observed in postmenopausal women with breast cancer. In this tumor growth study, dietary Avlimil at 500 ppm strongly stimulated MCF-7 tumor growth, but Avlimil at 1000 ppm had no apparent effect on the growth of MCF-7 tumors (Fig. 6). Increased pS2 expression was observed only in tumors from mice treated with 500 ppm Avlimil (Fig. 8). As seen *in vitro*, Avlimil apparently contains a mixture of both estrogenic and anti-estrogenic compounds with the net estrogenic actions occurring only at the lower dose. At 1000 ppm Avlimil it is likely that the anti-estrogenic component(s) dominated and no net estrogen-dependent tumor growth was observed. The different effect on tumor growth at different dietary dosage raised additional concerns regarding optimal dosage for either the purported efficacious effects of Avlimil or possible adverse outcomes.

Although Avlimil stimulated estrogen-dependent tumor growth and estrogen-responsive pS2 expression in mice, Avlimil (both at 500 and 1000 ppm) did not have an uterotrophic effect (Fig. 9). Increase in uterine weight is considered as a classic $ER\alpha$ -mediated estrogenic effect, and is typically used to measure estrogenicity of compounds (Cooke et al., 1997; Jefferson et al., 2002). Selective estrogen receptor modulators (SERMs) have been developed in order to control estrogenic effects on different target tissues. Tamoxifen, is anti-estrogenic in mammary and bone, but has an estrogen agonistic effect in the endometrium; conversely, Raloxifene, a second-generation SERM is also anti-estrogenic in mammary but has reduced agonistic effect on the endometrium (Zujewski, 2002). E_2 and tamoxifen treatment increase uterine weight in ovariectomized rats, but raloxifene and ICI had no effect (Stygar et al., 2003). E_2 treatment increased $ER\alpha$ mRNA expression in uterus, while tamoxifen increased $ER\beta$ mRNA expression (Stygar et al., 2003). The lack of effect by Avlimil on uterine weight is interesting and suggests that Avlimil or components in the herbal mixture have multiple biological activities and these activities may vary depending on the tissue. Additionally, the $ER\alpha/ER\beta$ -profile may also be a significant determinant of estrogenic activity in the various tissues. Because Avlimil is a complex mixture, its effects are likely to be derived from the combined actions of multiple bioactive compounds with estrogenic, anti-estrogenic, $ER\alpha$ -specific, or $ER\beta$ -specific mediated mechanisms. Additionally, it is possible that the mixture may behave as a SERM through individual bioactive compounds or the collective effect of dose of estrogenic and anti-estrogenic compounds may have different effects depending on the balance of these compounds. Since this herbal mixture contains complex estrogenic/anti-estrogenic activities as described

above it is very difficult to determine what the overall effect will be in specific tissues and individuals. These data suggest the individuals at risk for breast cancer should use caution before use of this complex product.

Initially we assumed that the isoflavone-containing herbal components in Avlimil (such as kudzu and red clover) would contribute most of the estrogenic activity; however, the isoflavones genistein, daidzein and equol were not detected in mouse blood ($<0.005 \mu\text{M}$) indicating that there was very little of these botanicals present. In 2004, the manufacturer introduced a new product 'Avlimil Complete' (Avlimil + 125 mg of 40% soy isoflavones concentrate) on the market claiming 'to restore a gentle normalization of estrogen levels at any stage of a woman's life' Avlimil Complete (2004). This dose would be approximately 63–188 ppm isoflavones based on the assumption that a woman consumes 0.5–1.5 kg food per day. Previously, we have demonstrated that the major soy isoflavone, genistein (125–1000 ppm), dose-dependently increased MCF-7 tumor growth (Ju et al., 2001), and soy products containing genistein (150 and 300 ppm) stimulated MCF-7 tumor growth (Allred et al., 2001). Therefore, the addition of estrogenic isoflavones to Avlimil may further enhance stimulation of estrogen-dependent breast cancer beyond that observed here. Further systematic studies on the interactions of the herbal mixture and isoflavones on breast cancer should be conducted before safe use can be assumed.

It is clear from this study that this estrogenic herbal mixture may put women at risk for adverse effects, especially if they have high risk factors for breast cancer. At this point, key bioactive components in Avlimil responsible for the estrogenic/anti-estrogenic activity and possible mechanisms of action are not known. Another critical observation was the fact that the lower dose of Avlimil (500 ppm) stimulated MCF-7 tumor growth *in vivo*, whereas the higher dose, 1000 ppm, did not. These data suggest that Avlimil may have multiple activities, estrogenic at lower concentrations and anti-estrogenic at higher concentrations. This is the first study that provides a systematic scientific evaluation of possible interactions between Avlimil and estrogen-dependent breast cancer. These observations, although provocative, require investigation of dosage effects of Avlimil in this pre-clinical postmenopausal breast cancer model and further identification of active constituents. A critical point of the research presented here is not that low doses stimulate tumor growth, but that only minor changes in dose can dramatically alter biological responses even though it is premature to state which dosage is safe or unsafe. The potential for adverse interactions of Avlimil on breast cancer suggest that further safety evaluations be conducted, particularly in light of the growing public enthusiasm for Avlimil supplementation.

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